

First degree.

Consider an equation $Y' + a(t) \cdot Y = 0$. Let's solve this.

If $Y(t_0) = 0$ for some $t_0 \in I$, then since $\tilde{Y} \equiv 0$ is a solution s.t. $\tilde{Y}(t_0) = 0$, the Uniqueness theorem gives $Y \equiv 0$. Now, assume $Y > 0$ for all $t \in I$.

Then, we obtain $\frac{Y'}{Y} = (\log Y)' = -a(x)$, so

$$\log Y = - \int_{t_0}^t a(x) dx \Rightarrow Y = \exp\left(- \int_{t_0}^t a(x) dx\right)$$

Non-Homogeneous Case

Consider an equation $Y' + a(t) \cdot Y = b(t)$.

We already know the Homogeneous Solution

$$Y_H = \exp\left(- \int_{t_0}^t a(x) dx\right)$$

And, the Particular Solution is

$$Y_p(t) = Y_H(t) \cdot \int_{t_0}^t \frac{b(x)}{Y_H(x)} dx$$

Therefore, the general Solution is

$$Y(t) = Y_H(t) \cdot \left[C + \int_{t_0}^t \frac{b(x)}{Y_H(x)} dx \right]$$

Chapter 3, section 3.

Consider the equation

$$y'' + \frac{1}{x} \cdot y' - \frac{1}{x^2} \cdot y = 0 \quad \text{for } x > 0.$$

Find the general solution of the ODE.

Proof. Observe that: let $y = x^r$ for some r . Then,

$$\begin{aligned} & r \cdot (r-1) \cdot x^{r-2} + \frac{1}{x} \cdot r \cdot x^{r-1} - \frac{1}{x^2} \cdot x^r \\ &= x^{r-2} (r(r-1) + r - 1) = x^{r-2} \cdot (r^2 - 1) \end{aligned}$$

Therefore, since $r^2 - 1 = (r-1)(r+1)$,

$y = x$ and $y = \frac{1}{x}$ are solutions,